

Weather Prediction Technical Report

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Abstract

Comparisons of New York and Los Angeles weather usually rest on personal preference rather than measurement. This project builds an automated pipeline that compares the two cities using defined metrics. An Airflow DAG retrieves daily weather data from the Open-Meteo API and loads it into Snowflake. A second DAG then runs dbt, which calculates derived metrics including rolling averages, temperature anomaly, dry-spell length, and comfortable-day counts. The results are displayed in a Preset dashboard. Over the period analyzed, Los Angeles averaged 7.57 comfortable days compared with 1.26 for New York. Its temperature stayed between roughly 72 and 81°F, while New York cooled from about 77°F to about 60°F. New York's temperature anomaly averaged 2.92 against 1.65 for Los Angeles. Los Angeles was therefore the more stable city, and New York showed more variation. Which city is preferable depends on whether a person favors steady conditions or change. The data covers only two cities over late summer into autumn, so the findings do not describe either city across a full year.

Keywords: weather, data pipeline, Airflow, dbt, Snowflake, Open-Meteo, Preset

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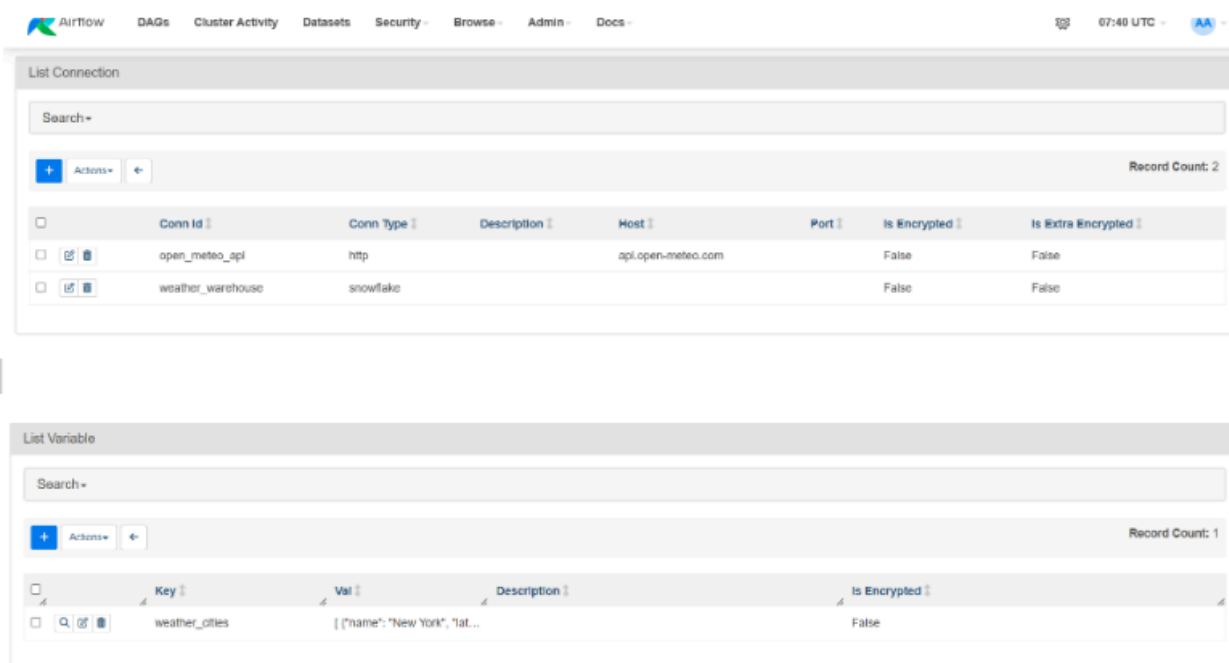
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The screenshot displays the Airflow web interface. At the top, there is a navigation bar with links for DAGs, Cluster Activity, Datasets, Security, Browse, Admin, and Docs. The main content area is divided into two sections: 'List Connection' and 'List Variable'.

List Connection

Search:

Actions:

Record Count: 2

☐	Conn Id	Conn Type	Description	Host	Port	Is Encrypted	Is Extra Encrypted
☐	open_meteo_api	http		api.open-meteo.com		False	False
☐	weather_warehouse	snowflake				False	False

List Variable

Search:

Actions:

Record Count: 1

☐	Key	Val	Description	Is Encrypted
☐	weather_cities	[{"name": "New York", "lat...}		False

Figure 1. Airflow connections and variable used by the pipeline.

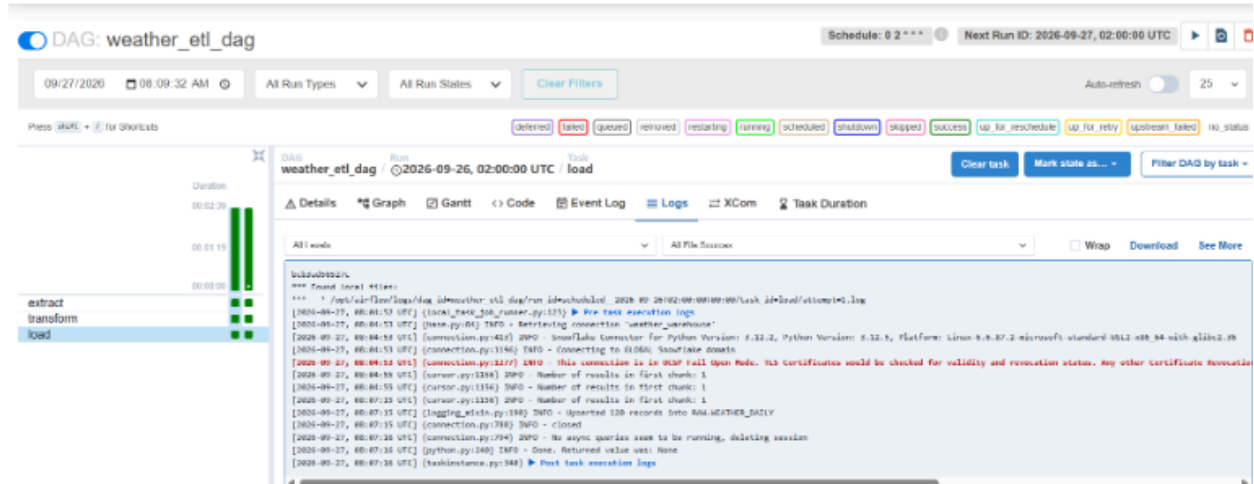


Figure 2. Airflow log for the weather_etl_dag load task.

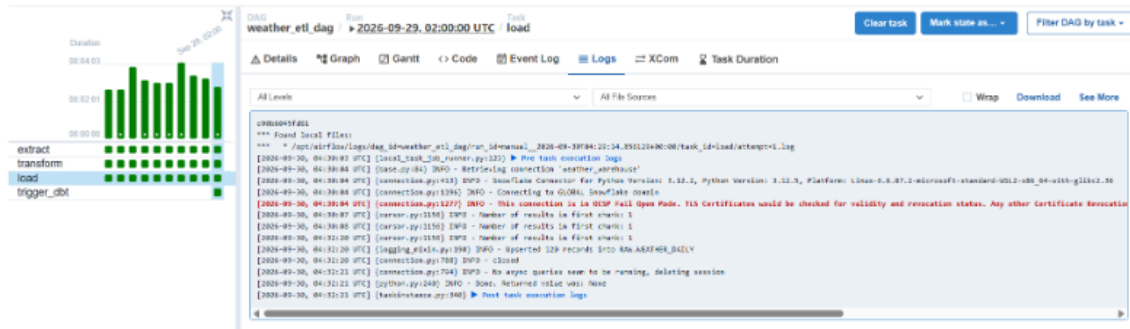
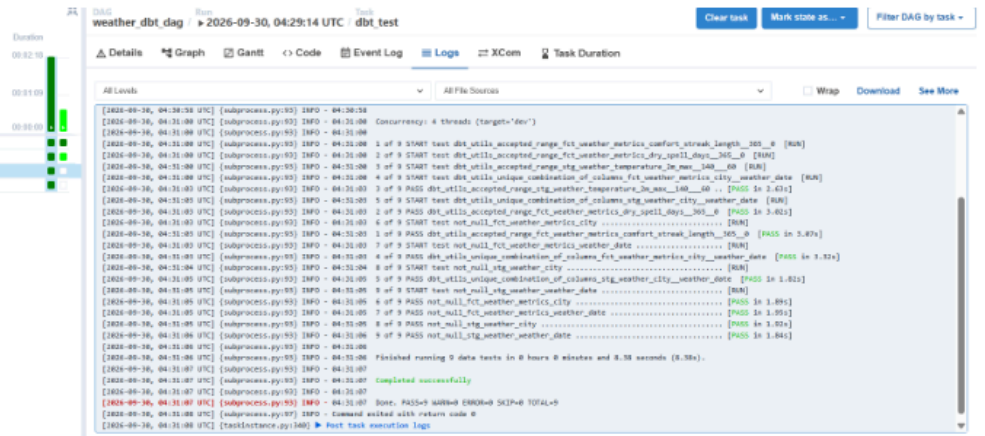


Figure 3. Airflow logs for the dbt_test and ETL load tasks.

Dbt run:

```

00:32:09 Running with dbt=1.8.7
00:32:11 Registered adapter: snowflake-1.8.1
00:32:15 Found 2 models, 1 snapshot, 9 data tests, 1 source, 570 macros
00:32:15 Concurrency: 4 threads (target='dev')
00:32:32 1 of 2 START sql view model ANALYTICS.stg_weather ..... [RUN]
00:32:36 1 of 2 OK created sql view model ANALYTICS.stg_weather ..... [SUCCESS 1 in 3.80s]
00:32:36 2 of 2 START sql table model ANALYTICS.fct_weather_metrics ..... [RUN]
00:32:40 2 of 2 OK created sql table model ANALYTICS.fct_weather_metrics ..... [SUCCESS 1 in 4.51s]
00:32:40 Finished running 1 view model, 1 table model in 0 hours 0 minutes and 25.70 seconds (25.70s).
00:32:40 Completed successfully
00:32:40 Done, PASS=2 WARN=0 ERROR=0 SKIP=0 TOTAL=2
00:32:47 Running with dbt=1.8.7

```

Dbt test:

```

00:32:47 Running with dbt=1.8.7
00:32:49 Registered adapter: snowflake-1.8.1
00:32:51 Found 2 models, 1 snapshot, 9 data tests, 1 source, 570 macros
00:32:51 Concurrency: 4 threads (target='dev')
00:32:58 1 of 9 START test dbt_utils_accepted_range_fct_weather_metrics_comfort_streak_length_60_0 [RUN]
00:32:58 2 of 9 START test dbt_utils_accepted_range_fct_weather_metrics_dry_spell_days_60_0 [RUN]
00:32:58 3 of 9 START test dbt_utils_accepted_range_stg_weather_temperature_20_max_140_60 [RUN]
00:32:58 4 of 9 START test dbt_utils_unique_combination_of_columns_fct_weather_metrics_city_weather_date [RUN]
00:33:01 1 of 9 PASS dbt_utils_accepted_range_fct_weather_metrics_comfort_streak_length_60_0 [PASS in 3.41s]
00:33:01 5 of 9 START test dbt_utils_unique_combination_of_columns_stg_weather_city_weather_date [RUN]
00:33:03 2 of 9 PASS dbt_utils_accepted_range_fct_weather_metrics_dry_spell_days_60_0 [PASS in 5.09s]
00:33:03 6 of 9 START test not_null_fct_weather_metrics_city ..... [RUN]
00:33:07 5 of 9 PASS dbt_utils_unique_combination_of_columns_stg_weather_city_weather_date [PASS in 5.89s]
00:33:07 7 of 9 START test not_null_fct_weather_metrics_weather_date ..... [RUN]
00:33:08 4 of 9 PASS dbt_utils_unique_combination_of_columns_fct_weather_metrics_city_weather_date [PASS in 9.53s]
00:33:08 8 of 9 START test not_null_stg_weather_city ..... [RUN]
00:33:09 8 of 9 PASS not_null_stg_weather_city ..... [PASS in 1.79s]
00:33:09 9 of 9 START test not_null_stg_weather_weather_date ..... [RUN]
00:33:10 7 of 9 PASS not_null_fct_weather_metrics_weather_date ..... [PASS in 2.32s]
00:33:10 3 of 9 PASS dbt_utils_accepted_range_stg_weather_temperature_20_max_140_60 .. [PASS in 12.09s]
00:33:10 6 of 9 PASS not_null_fct_weather_metrics_city ..... [PASS in 7.11s]
00:33:14 9 of 9 PASS not_null_stg_weather_weather_date ..... [PASS in 4.29s]
00:33:14 Finished running 9 data tests in 0 hours 0 minutes and 22.26 seconds (22.26s).
00:33:14 Completed successfully
00:33:14 Done, PASS=9 WARN=0 ERROR=0 SKIP=0 TOTAL=9

```

Dbt snapshot:

```

00:33:22 Running with dbt=1.8.7
00:33:24 Registered adapter: snowflake-1.8.1
00:33:27 Found 2 models, 1 snapshot, 9 data tests, 1 source, 570 macros
00:33:27 Concurrency: 4 threads (target='dev')
00:33:39 1 of 1 START snapshot snapshots.raw_weather_snapshot ..... [RUN]
00:33:42 1 of 1 OK snapshoted snapshots.raw_weather_snapshot ..... [SUCCESS 1 in 3.67s]
00:33:42 Finished running 1 snapshot in 0 hours 0 minutes and 15.15 seconds (15.15s).
00:33:43 Completed successfully
00:33:43 Done, PASS=1 WARN=0 ERROR=0 SKIP=0 TOTAL=1

```

Figure 4. Terminal output for dbt run, dbt test and dbt snapshot.

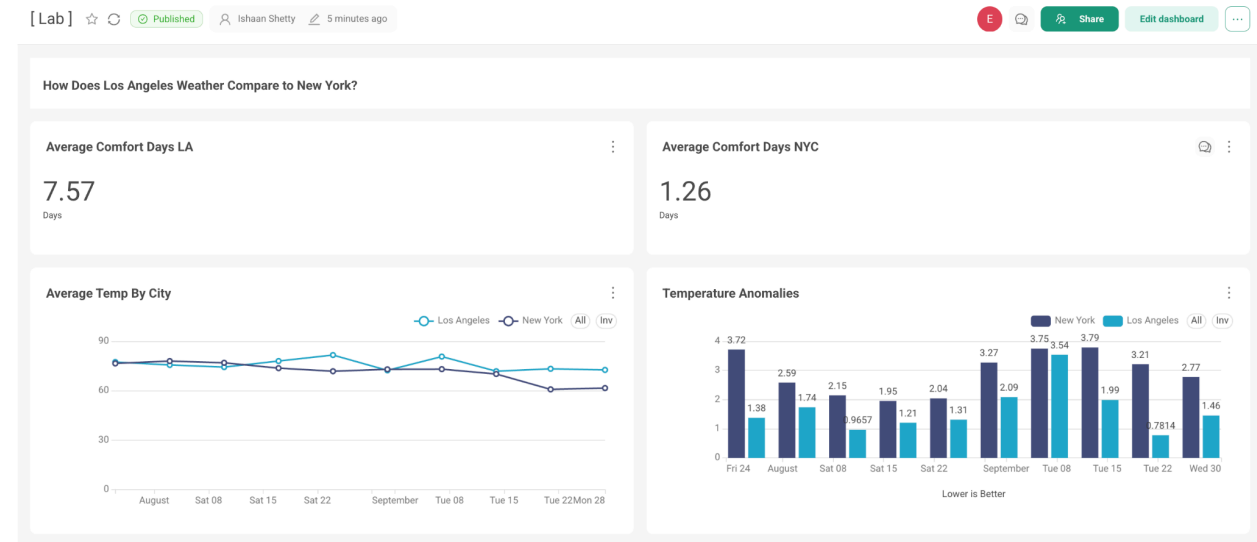


Figure 5. Preset dashboard: comfort days, temperature and anomalies.

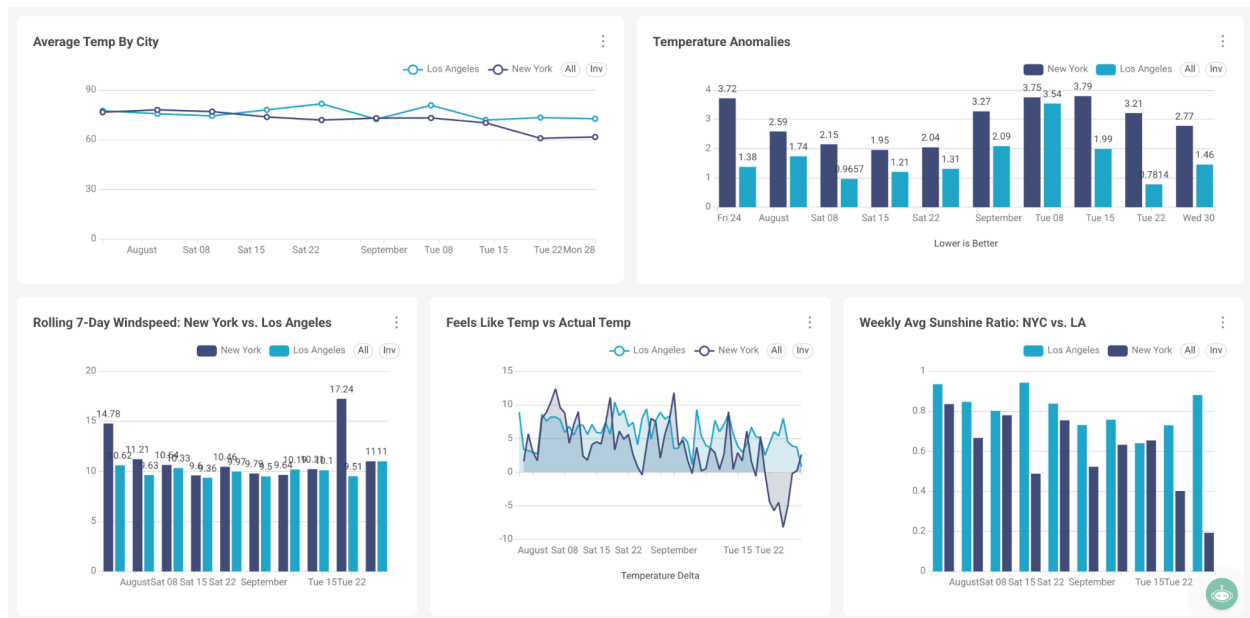


Figure 6. Preset dashboard: temperature, wind, feels-like and sunshine charts.

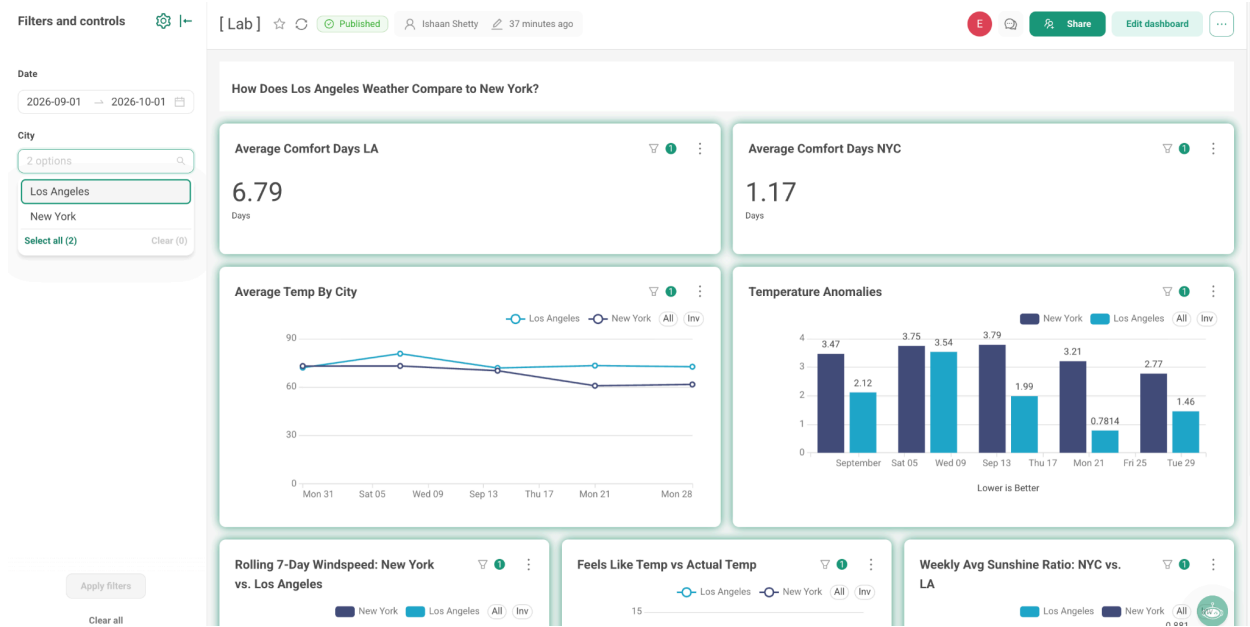


Figure 7. Preset dashboard in edit mode with date and city filters.

List of Tables

Table 1. DDL for the raw table RAW.WEATHER_DAILY.

```
CREATE TABLE IF NOT EXISTS RAW.WEATHER_DAILY (
  CITY                VARCHAR(50)    NOT NULL,
  DATE                DATE            NOT NULL,
  LATITUDE            NUMBER(9, 6)    NOT NULL,
  LONGITUDE           NUMBER(9, 6)    NOT NULL,
  TEMPERATURE_2M_MAX  NUMBER(5, 1),
  TEMPERATURE_2M_MIN  NUMBER(5, 1),
  TEMPERATURE_2M_MEAN NUMBER(5, 1),
  APPARENT_TEMPERATURE_MAX NUMBER(5, 1),
  PRECIPITATION_SUM   NUMBER(6, 3),
  PRECIPITATION_HOURS NUMBER(4, 1),
  PRECIPITATION_PROBABILITY_MAX NUMBER(3, 0),
  WIND_SPEED_10M_MAX  NUMBER(5, 1),
  SUNSHINE_DURATION   NUMBER(8, 1),
  DAYLIGHT_DURATION   NUMBER(8, 1),
  UV_INDEX_MAX        NUMBER(4, 2),
  WEATHER_CODE         NUMBER(2, 0),
  RETRIEVED_AT        TIMESTAMP_NTZ NOT NULL DEFAULT CURRENT_TIMESTAMP(),
  PRIMARY KEY (CITY, DATE)
);
```

Table 2. DDL for the mart table ANALYTICS.FCT_WEATHER_METRICS.

```

CREATE TABLE IF NOT EXISTS ANALYTICS.FCT_WEATHER_METRICS (
  CITY                                VARCHAR(50)    NOT NULL,
  WEATHER_DATE                       DATE          NOT NULL,
  LATITUDE                           NUMBER(9,6),
  LONGITUDE                           NUMBER(9,6),

  -- passthrough raw metrics
  TEMPERATURE_2M_MAX                 NUMBER(5,1),
  TEMPERATURE_2M_MIN                 NUMBER(5,1),
  TEMPERATURE_2M_MEAN                 NUMBER(5,1),
  APPARENT_TEMPERATURE_MAX            NUMBER(5,1),
  PRECIPITATION_SUM                   NUMBER(6,3),
  WIND_SPEED_10M_MAX                  NUMBER(5,1),
  UV_INDEX_MAX                        NUMBER(4,2),
  WEATHER_CODE                        NUMBER(2,0),

  -- derived metrics
  WEATHER_WHIPLASH_INDEX               NUMBER(5,2),
  PREDICTABILITY_SCORE                 NUMBER(5,3),
  ROLLING_TEMP_AVG_7D                  NUMBER(5,1),
  ROLLING_RAINFALL_TOTAL_7D            NUMBER(6,3),
  TEMP_ANOMALY_VS_BASELINE             NUMBER(5,1),
  DRY_SPELL_DAYS                       NUMBER(3,0),
  COMFORT_STREAK_LENGTH                NUMBER(3,0),
  IS_LAYER_UP_DAY                      BOOLEAN,
  FEELS LIKE GAP                       NUMBER(5,1),
  RAIN_INTENSITY                       NUMBER(6,3),
  SUNSHINE_RATIO                       NUMBER(5,3),
  IS_WEEKEND_RAIN_DAY                  BOOLEAN,
  IS_UV_DANGER_DAY                     BOOLEAN,

  PRIMARY KEY (CITY, WEATHER_DATE)
);

```

Nomenclature and Abbreviations

#	Metric	Definition / Calculation	Source Fields	Purpose / Business Question
1	Preference-based head-to-head win rate	Score each city daily against user-selected thresholds (temperature range, rain tolerance, wind). Count the days each city wins.	temperature_2m_max/min, precipitation_sum, wind_speed_10m_max	Directly answers "which coast?" and updates with dashboard filters
2	Weather whiplash index	Absolute difference between today's and yesterday's mean temperature, averaged over a rolling 7 days	temperature_2m_mean	Measures how erratic each city's weather feels
3	Predictability score	Coefficient of variation (std dev ÷ mean) of daily temperature over a window	temperature_2m_mean	Separates people who want stable weather from those who want variety
4	Layer-up day flag	Flag days where max temp minus min temp exceeds about 20°F	temperature_2m_max, temperature_2m_min	Practical view of diurnal range
5	Feels-like gap	Apparent temperature max minus actual temperature max	apparent_temperature_max, temperature_2m_max	Shows how humidity and wind change perceived temperature
6	Rain intensity	Precipitation sum ÷ precipitation hours on wet days	precipitation_sum, precipitation_hours	Distinguishes constant drizzle from brief downpours
7	Sunshine ratio	Sunshine duration ÷ daylight duration	sunshine_duration, daylight_duration	"Gloom index" showing the share of possible sun each city gets
8	Weekend ruined rate	Share of Saturdays and Sundays with precipitation above a threshold, compared to weekdays	precipitation_sum, date	Relatable metric with a clear takeaway
11	Comfort streak length	Longest run of consecutive days meeting the "comfortable" criteria	temperature_2m_mean, precipitation_sum, wind_speed_10m_max	Good-weather counterpart to dry spell length
12	UV danger days	Count of days with UV index of 8 or higher	uv_index_max	Adds a health angle

1. Introduction

1.1 Background

People like to argue about whether New York or Los Angeles has better weather, and the argument usually comes down to opinion. One person cares about sunshine, another hates humidity, and a third just wants a week where they don't have to check the forecast. Without agreed measures, nobody wins the argument.

For this project we picked a set of measurable questions instead. How many days are comfortable in each city? How long do dry stretches last? How much does the temperature jump from one day to the next? How steady is the weather over a week? Each of these can be calculated from daily weather readings, so the two cities can be compared on the same terms.

To get those readings we used the Open-Meteo API. We send it the latitude and longitude of each city and get back daily values for temperature, precipitation, wind, sunshine and UV index. Both cities go through the same request and the same calculations, so any difference in the results comes from the weather and not from how the data was handled.

We also wanted the comparison to stay current without anyone rerunning it by hand. The pipeline therefore pulls new data every day on a schedule and updates the metrics automatically, so the results always reflect the latest readings.

1.2 Problem Statement

Debates over whether New York or Los Angeles has better weather usually rest on personal preference and memory rather than measurement. This project addresses that gap by collecting daily weather data for both cities and calculating a consistent set of metrics, including comfortable-day counts, sunshine days, and temperature variability/predictability. The goal is to give a measured, repeatable answer to which city's weather is better.

1.3 Scope and Limitations

Scope

1. Retrieve daily weather data for two cities, New York and Los Angeles, from the Open-Meteo API on daily schedule, and load it into a Snowflake table.
2. Use dbt to transform the raw readings into derived metrics, including moving averages, temperature anomaly, rolling rainfall, dry spell length, and comfort streak length.
3. Use a BI tool (Preset) to visualize the metrics from step 2 in a dashboard where a viewer can filter by city and date range.

Limitations

1. **Two cities only:** Results describe New York and Los Angeles and should not be read as a comparison of the East and West Coasts as a whole.
2. **Short time window:** Each run loads the most recent 60 days of data, so the analysis cannot show seasonal patterns or long term trends. A full year of data would be needed to compare a complete seasonal cycle.
3. **Daily resolution:** The data holds one reading per city per day, so it cannot capture conditions within a day, such as a morning rain shower or an afternoon heat spike.
4. **Observed data, no forecasts:** The pipeline records past daily values and does not store forecasts. Metrics that compare forecast with actual weather, such as forecast miss rate, are outside this project.
5. **Chosen thresholds:** Metrics like dry day (under 0.01 inches of rain) and comfortable day (65 to 80°F mean temperature, no rain, wind under 15 mph) rest on our own definitions. A different definition of "comfortable" could change which city looks better.
6. **Window-based metrics:** Rolling averages and anomalies are calculated over trailing 7- and 30-day windows, so the first days of each series are based on fewer observations.

2. Methodology

2.1 Approach Overview

The pipeline was designed to satisfy the project requirements, which specified retrieving weather data for two cities from the Open-Meteo API, transforming it with dbt, and visualizing the results in a business intelligence tool. Each requirement corresponds to one stage of the pipeline.

An Airflow DAG executes daily, retrieves the data for New York and Los Angeles, and loads it into Snowflake. A second DAG then runs dbt, which cleans the raw data and calculates the derived metrics. Preset reads the resulting table to populate the dashboard (Figure 8).

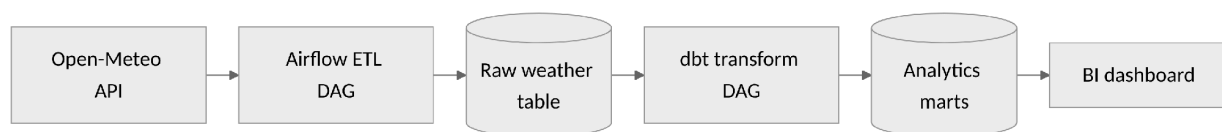


Figure 8. Pipeline architecture from the Open-Meteo API to the dashboard.

The stages were kept separate so that each could be developed and tested independently without affecting the others.

2.2 Materials, Equipment, and Software

Component	Choice
Data source	Open-Meteo Forecast API (https://api.open-meteo.com/v1/forecast)
Cities	New York, Los Angeles
Orchestration	Airflow
Warehouse	Snowflake
Transformation	Dbt (python,yaml)
BI tool	Preset
Version control	GitHub, one repo

2.3 Assumptions and Constraints

2.3.1 Assumptions

- We took the Open-Meteo data as accurate and did not check it against another weather source. Each city is represented by a single latitude and longitude, which treats one point as the weather for the whole city. We also assumed that 60 days of data is enough to compare the two cities, and that one reading per day is detailed enough for the metrics we calculate.
- The dry-day and comfortable-day definitions are our own choices. A day counts as dry if it has less than 0.01 inches of precipitation. A day counts as comfortable if the mean temperature is between 65 and 80°F, there is no measurable rain, and wind stays under 15 mph. Other people might define a good day differently, and the results depend on these cutoffs.

2.3.2 Constraints

- The project requirements fixed the main tools: the Open-Meteo API for data, dbt for transformation, and a BI tool for visualization. They also limited the work to two cities, and the API request returns only the last 60 days of daily values.
- Loading data and running dbt both need a Snowflake account. The connection uses a private key that is kept out of the repository, so each person running the pipeline has to get their own copy. The pipeline runs in a local Docker setup, so it only runs while that environment is up. Access to edit and share the dashboard depends on the permissions set in the BI tool.

2.4 Procedure

1. **Environment Setup:** Set up a Docker Compose file running Airflow, installed Snowflake and dbt packages via `_PIP_ADDITIONAL_REQUIREMENTS`, and mounted a `keys/` folder for authentication.
2. **Snowflake Key-Pair Auth:** Used RSA key-pair authentication, with the private key mounted into the container and its passphrase stored in a gitignored `.env` file.

3. **Airflow Connections and Variables:** Created open_meteo_api (HTTP), weather_warehouse (Snowflake, key-pair auth), and a weather_cities Variable holding NY/LA coordinates.
4. **ETL DAG:** Built weather_etl_dag.py with extract → transform → load tasks that pull daily weather from Open-Meteo and upsert it into RAW.WEATHER_DAILY via a Snowflake MERGE, ensuring idempotency.
5. **dbt Project:** Built a staging model (stg_weather) cleaning the raw table, and a mart (fct_weather_metrics) computing 10 derived metrics using window functions, both with schema tests.
6. **Snapshot:** Added raw_weather_snapshot.sql using the timestamp strategy to track changes to the raw table over time.
7. **dbt DAG:** Built weather_dbt_dag.py, chaining dbt deps → dbt run → dbt test → dbt snapshot
8. **BI Dashboard:** Connected Preset to Snowflake via username/password auth and built a dashboard on fct_weather_metrics with temperature, wind, sunshine

2.5 Data Collection and Analysis

2.5.1 Data Collection

Daily weather data was retrieved from the Open-Meteo forecast endpoint for New York and Los Angeles, using the latitude and longitude of each city. Each request covered the most recent 60 days and returned twelve daily variables: maximum, minimum and mean temperature, maximum apparent temperature, precipitation total, precipitation hours, maximum precipitation probability, maximum wind speed, sunshine duration, daylight duration, maximum UV index, and weather code. Temperature was requested in degrees Fahrenheit, precipitation in inches, and wind speed in miles per hour, with dates reported in each city's local time zone. The scheduled run produced one record per city per day, for a total of 120 records per load.

Records were loaded into a raw Snowflake table keyed on city and date. A merge operation inserted new records and updated existing ones only when a value had changed, so repeated runs did not create duplicates. The load ran as a single transaction and was rolled back if any step failed.

2.5.2 Data Analysis

Analysis was performed in dbt in two layers. The staging model renamed fields and converted sunshine and daylight duration from seconds to hours. The mart model then calculated the derived metrics separately for each city and day, so that the two series can be compared directly.

Trend and variability were measured with rolling windows. A 7-day rolling average of daily mean temperature and a 7-day rolling rainfall total smoothed out day-to-day noise. Temperature anomaly was calculated as the difference between a day's mean temperature and the average over a trailing 30-day window, showing how far each day departed from recent conditions. Two further measures described stability. The weather whiplash index averaged the absolute day-to-day change in mean temperature over 7 days, and the predictability score divided the standard deviation of daily temperature by its mean over the same window, so that lower values indicate steadier weather.

Streak metrics described how long favorable or unfavorable conditions lasted. A day was classified as dry if precipitation was below 0.01 inches, and as comfortable if the mean temperature was between 65 and 80°F, precipitation was below 0.01 inches, and wind speed was under 15 miles per hour. Dry-spell length and comfort-streak length counted the consecutive days meeting each condition and reset to zero when a day did not.

3. Results

The dashboard time axis runs from late July to September 30, 2026. All values below are from that range. Figures 5 to 7 show the full dashboard, and Figures 9 to 14 show each chart on its own.

3.1 Comfortable Days

Los Angeles averaged 7.57 comfortable days and New York averaged 1.26 (Figure 9). A comfortable day was one with a mean temperature of 65 to 80°F, less than 0.01 in of precipitation, and wind under 15 mph.

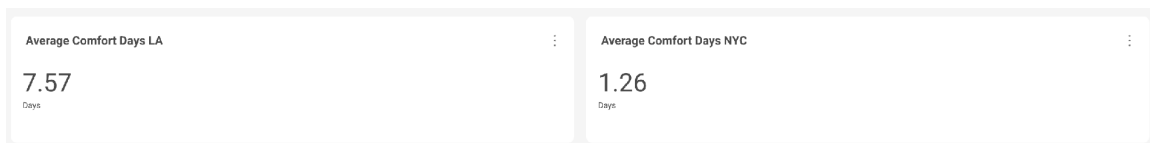


Figure 9. Average comfort days, Los Angeles and New York.

3.2 Temperature

Both cities started the period at similar average temperatures, near 77°F (Figure 10). Los Angeles then stayed between roughly 72 and 81°F for the rest of the period. New York declined from early September and was near 60°F at the end of the period.

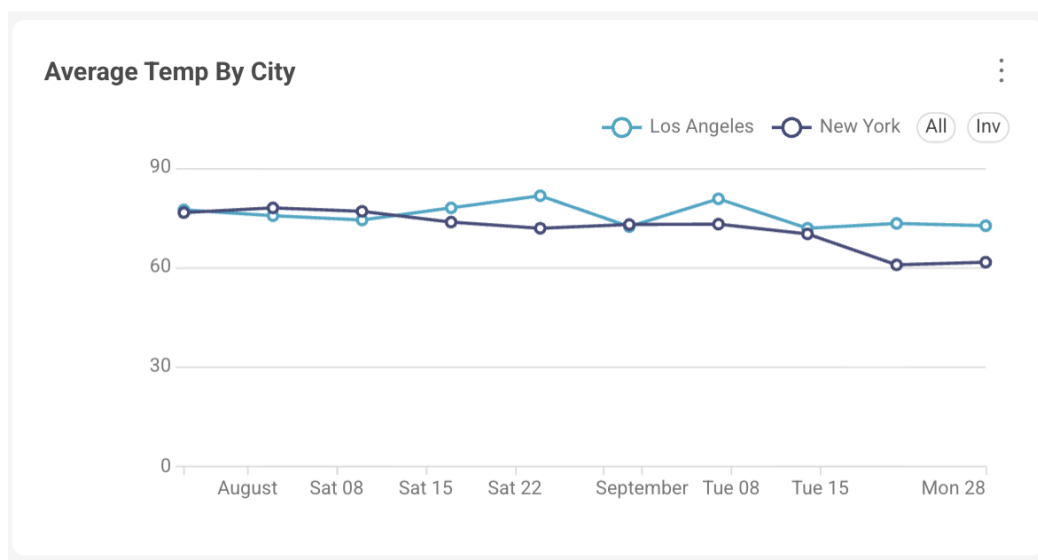


Figure 10. Average temperature by city.

3.3 Temperature Anomaly

New York's temperature anomaly was higher than Los Angeles's in all ten periods plotted (Figure 11). The plotted values averaged 2.92 for New York and 1.65 for Los Angeles. New York's values ranged from 1.95 to 3.79, and Los Angeles's from 0.78 to 3.54. The two cities were closest in the period beginning Tuesday, September 8 (3.75 and 3.54).

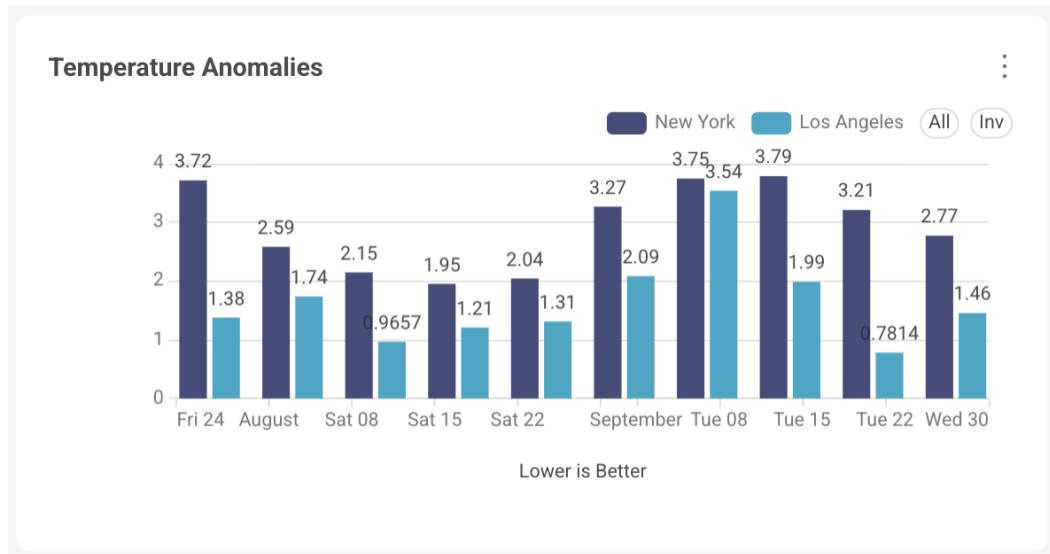


Figure 11. Temperature anomalies by city (lower is better).

3.4 Wind

Rolling 7-day wind speed was between about 9 and 11 mph for both cities in most periods (Figure 12). New York recorded two higher readings, 14.78 and 17.24 mph.

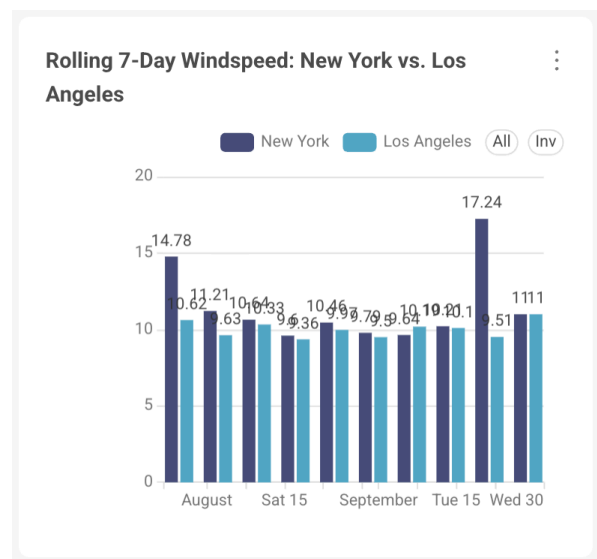


Figure 12. Rolling 7-day wind speed, New York vs. Los Angeles.

3.5 Feels-Like Difference

The difference between feels-like and actual temperature stayed between about 0 and 10 degrees for both cities through most of the period (Figure 13). Near the end of the period, New York's difference fell below zero, reaching about -8.

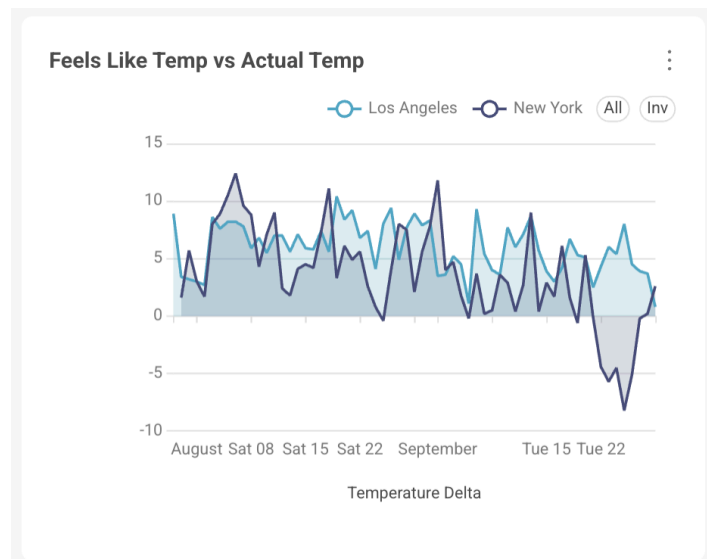


Figure 13. Feels-like temperature vs. actual temperature.

3.6 Sunshine

Los Angeles had the higher weekly sunshine ratio in most weeks, generally between about 0.7 and 0.9 (Figure 14). New York's ratio was lower and varied more from week to week, and was about 0.2 in the final week.

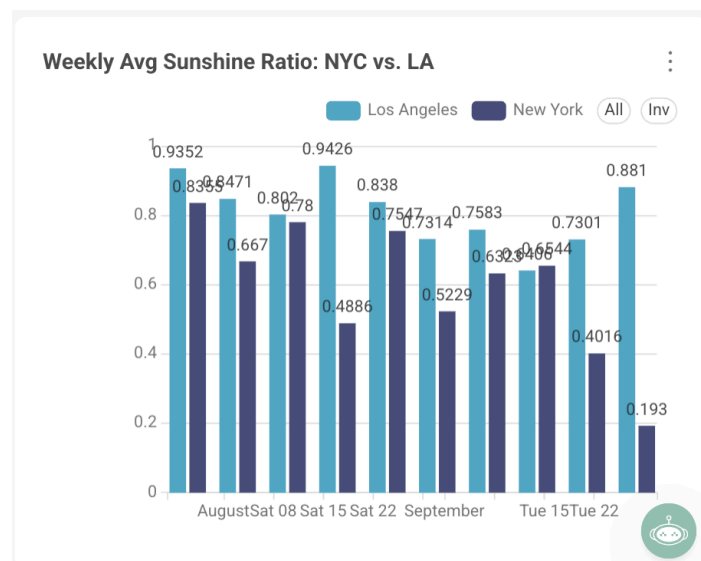


Figure 14. Weekly average sunshine ratio, NYC vs. LA.

4. Conclusions

This project built an automated pipeline that collects daily weather for New York and Los Angeles, calculates comparison metrics in dbt, and displays them in a Preset dashboard. Over the period analyzed, Los Angeles had the more stable weather. It averaged 7.57 comfortable days against 1.26 for New York, and its temperature stayed within a narrow band of roughly 72 to 81°F. New York cooled from about 77°F to about 60°F over the same period. Its temperature anomaly averaged 2.92 compared with 1.65 for Los Angeles, which means its temperatures moved further from recent conditions. Los Angeles also had the higher sunshine ratio in most weeks.

The answer to which city has better weather depends on what a person wants from it. For someone who prefers steady, mild, sunny conditions, Los Angeles is the better choice by every measure we calculated. For someone who prefers change from week to week, New York is the better choice, since its temperatures, wind, and sunshine varied much more. The metrics measure stability, so they cannot say which kind of weather a person will enjoy more.

5. Future Work

For future iterations, the pipeline could be extended to support additional cities through a fully config-driven list rather than hardcoded entries, and could incorporate data quality alerting, such as a Slack notification on dbt test failures, to catch issues proactively. Backfilling historical weather via Open-Meteo's archive API would allow for a longer anomaly baseline, improving the accuracy of temperature anomaly calculations. A forecast-accuracy metric comparing forecasted values against observed outcomes would add further analytical depth, and migrating from Postgres to a cloud warehouse would support larger-scale, multi-city expansion down the line.

References

- [1] Open-Meteo, "Weather Forecast API," Open-Meteo.com. [Online]. Available: <https://api.open-meteo.com/v1/forecast>. [Accessed: Sep. 30, 2026].